HU Collection

TMUA Mock Set A, Paper 1

Written by Kevin Hu

Recommended time:	2 hours (the live TMUA allows 75 minutes)
Calculators:	not permitted
Format:	20 multiple-choice questions

This is an original mock paper written in the style of Paper 1 of the Test of Mathematics for University Admission (TMUA), the admissions test used for Mathematics, Economics and Computer Science courses at universities including Cambridge, Oxford, Imperial College London, Warwick, Durham, LSE and UCL. Paper 1 tests the application of mathematical knowledge: algebra, functions, sequences and series, coordinate geometry, trigonometry, logarithms, differentiation, integration, number, probability and statistics.

Guidance

The questions here sit at the hard end of the TMUA range. Some of them will take longer than the four minutes or so that the live test allows per question, and one or two contain a trap that catches most people on a first attempt. That is deliberate. The purpose of this paper is practice rather than measurement, so treat a wrong answer as the start of a lesson rather than the end of one.

Each question has exactly one correct option. No marks are lost for a wrong answer, so every question should be attempted. As a rough benchmark, 14 or more correct answers on this paper suggests you are already comfortable with the hardest material the live test is likely to contain.

Answers and full worked solutions are published separately on the HU Collection website. If you spot an error, or find a neater method, I would be glad to hear from you.

Good luck.

Kevinvariant
kevinvariant08.github.io

Question 1

Find the sum of all real solutions of the equation

$$\left(x^2 - 5x + 5\right)^{x^2-9x+20} = 1.$$

A 5

B 9

C 10

D 12

E 15

F 16

Kevinvariant
kevinvariant08.github.io

Question 2

How many positive divisors of $2^6 \times 3^4$ are perfect squares but not perfect cubes?

A 8

B 10

C 12

D 14

E 24

F 35

Kevinvariant
kevinvariant08.github.io

Question 3

Find the complete set of values of k for which

$$-3 < \frac{x^2 + kx + 1}{x^2 + x + 1} < 3$$

for every real number x.

A $-1 < k < 7$

B $-11 < k < 5$

C $-11 < k < 7$

D $-5 < k < 1$

E $-1 < k < 5$

F all real values of k

Kevinvariant
kevinvariant08.github.io

Question 4

A sequence is defined by $u_1 = 2$ and

$$u_{n+1} = \frac{u_n - 1}{u_n + 1} \qquad \text{for } n \geq 1.$$

Find the value of $\sum_{n=1}^{100} u_n$.

A $-\frac{175}{6}$

B $-\frac{50}{3}$

C $-\frac{29}{6}$

D $-\frac{7}{6}$

E $\frac{50}{3}$

F $\frac{175}{6}$

Kevinvariant
kevinvariant08.github.io

Question 5

The real numbers a and b satisfy $a > b > 1$, $ab = 256$ and

$$\log_a b + \log_b a = \frac{10}{3}.$$

Find the value of $a - b$.

A -60

B 12

C 48

D 60

E 62

F 252

Kevinvariant
kevinvariant08.github.io

Question 6

How many solutions does the equation

$$\sin^2 x + \sin^2 2x = \cos^2 x + \cos^2 2x$$

have in the interval $0° \leq x < 360°$?

A 2

B 4

C 5

D 6

E 8

F 12

Kevinvariant
kevinvariant08.github.io

Question 7

A cubic polynomial $p(x)$ satisfies

$$p(1) = 1, \qquad p(2) = 2, \qquad p(3) = 3, \qquad p(4) = 5.$$

Find the value of $p(5)$.

A 7

B 8

C 9

D 10

E 11

F 12

Kevinvariant
kevinvariant08.github.io

Question 8

How many real solutions does the equation

$$\left|x^2 - 2x\right| = |x - 1|$$

have?

A 0

B 1

C 2

D 3

E 4

F 6

Kevinvariant
kevinvariant08.github.io

Question 9

Three fair six-sided dice are rolled. Given that the total score is 10, find the probability that the three scores are all different.

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{5}{9}$

D $\frac{2}{3}$

E $\frac{3}{4}$

F $\frac{5}{6}$

Kevinvariant
kevinvariant08.github.io

Question 10

Find the coefficient of x^2 in the expansion of

$$(1+x)^2 + (1+x)^3 + (1+x)^4 + \cdots + (1+x)^{20}.$$

A 1140

B 1330

C 1540

D 1771

E 2870

F 3080

Kevinvariant
kevinvariant08.github.io

Question 11

The normal to the curve $y = x^2$ at the point $P(a, a^2)$, where $a > 0$, meets the curve again at the point Q.

Find the least possible value of the y-coordinate of Q as a varies.

A $\sqrt{2}$

B $\frac{3}{2}$

C 2

D $\frac{9}{4}$

E $2\sqrt{2}$

F 4

Kevinvariant
kevinvariant08.github.io

Question 12

Find the area of the finite region enclosed by the line $y = 5$ and the curve $y = \left|x^2 - 4\right|$.

A 14

B $\frac{46}{3}$

C 16

D $\frac{52}{3}$

E $\frac{74}{3}$

F $\frac{44}{3}$

Kevinvariant
kevinvariant08.github.io

Question 13

A data set consists of eight numbers with mean 5 and variance 4. Two of the numbers, 1 and 9, are removed.

Find the variance of the remaining six numbers.

A 0

B $\frac{4}{3}$

C 2

D $\frac{8}{3}$

E 4

F $\frac{16}{3}$

G It cannot be determined from the information given.

Kevinvariant
kevinvariant08.github.io

Question 14

How many real numbers x satisfy

$$\sqrt{x+3-4\sqrt{x-1}}+\sqrt{x+8-6\sqrt{x-1}}=1\ ?$$

A 0

B 1

C 2

D 3

E 4

F infinitely many

Kevinvariant
kevinvariant08.github.io

Question 15

Let $\lfloor t \rfloor$ denote the largest integer that is less than or equal to t. Find the value of

$$\sum_{n=1}^{100} \lfloor \sqrt{n} \rfloor .$$

A 615

B 625

C 635

D 645

E 655

F 671

Kevinvariant
kevinvariant08.github.io

Question 16

Find the maximum value of

$$\frac{\sin x}{2+\cos x}$$

as x ranges over all real numbers.

A $\frac{1}{4}$

B $\frac{1}{3}$

C $\frac{1}{2}$

D $\frac{\sqrt{3}}{3}$

E $\frac{\sqrt{2}}{2}$

F $\sqrt{3}$

Kevinvariant
kevinvariant08.github.io

Question 17

Find the complete set of values of x in the interval $0 \leq x \leq 2\pi$ for which

$$\sin x \geq \cos 2x.$$

A $\frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$

B $\frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$ or $x = \frac{3\pi}{2}$

C $0 \leq x \leq \frac{\pi}{6}$ or $\frac{5\pi}{6} \leq x \leq 2\pi$

D $\frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$ or $\frac{7\pi}{6} \leq x \leq \frac{11\pi}{6}$

E $\frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$ or $x = \frac{7\pi}{6}$ or $x = \frac{11\pi}{6}$

F $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$ or $x = \frac{3\pi}{2}$

Kevinvariant
kevinvariant08.github.io

Question 18

The circle C has equation $x^2 + y^2 = 4$, and O is the origin. A straight line through the point $P(1, 0)$ meets C at two points A and B.

Find the maximum possible area of triangle OAB.

A 1

B $\frac{3}{2}$

C $\sqrt{3}$

D 2

E $2\sqrt{3}$

F 4

Kevinvariant
kevinvariant08.github.io

Question 19

A straight line ℓ is a tangent to the curve

$$y = x^4 - 4x^3 + 2x^2$$

at two distinct points. Find the gradient of ℓ.

A -4

B -8

C -2

D -1

E 2

F 4

G No such line exists.

Kevinvariant
kevinvariant08.github.io

Question 20

The function f is defined by $f(x) = x^3 + x$ for $x \geq 0$, and f^{-1} denotes its inverse function. Find the value of

$$\int_0^{10} f^{-1}(x)\,dx.$$

A 6

B 8

C 10

D 12

E 14

F 20

Kevinvariant
kevinvariant08.github.io

Answer key

Q	1	2	3	4	5	6	7	8	9	10
	E	B	E	A	D	D	C	E	D	B

Q	11	12	13	14	15	16	17	18	19	20
	C	F	A	F	B	D	B	C	A	E

Worked solutions

Question 1

A power equals 1 in three ways: the base is 1, or the base is -1 and the exponent is an even integer, or the exponent is 0 and the base is non-zero. Base 1 means $x^2-5x+4=0$, giving $x=1$ or $x=4$. Base -1 means $x^2-5x+6=0$, giving $x=2$ or $x=3$; the exponents there are 6 and 2, both even, so both count. Exponent 0 means $x^2-9x+20=0$, giving $x=4$ or $x=5$, and the bases 1 and 5 are non-zero. The base is never zero at any of these points. The solutions are $1,2,3,4,5$ with sum 15. **Answer E.**

Question 2

A divisor has the form 2^a3^b with $0\le a\le 6$ and $0\le b\le 4$. It is a square when both exponents are even, so $a\in\{0,2,4,6\}$ and $b\in\{0,2,4\}$, giving 12 squares. It is also a cube when both exponents are multiples of 6, so $a\in\{0,6\}$ and $b=0$, giving 2 sixth powers. That leaves $12-2=10$. **Answer B.**

Question 3

The denominator x^2+x+1 is positive for all x, so both inequalities can be multiplied out safely. The right-hand inequality becomes $2x^2+(3-k)x+2>0$ for all x, which needs discriminant $(3-k)^2-16<0$, so $-1<k<7$. The left-hand inequality becomes $4x^2+(k+3)x+4>0$ for all x, which needs $(k+3)^2-64<0$, so $-11<k<5$. Both must hold, giving $-1<k<5$. **Answer E.**

Question 4

Computing directly, $u_1=2$, $u_2=\frac{1}{3}$, $u_3=-\frac{1}{2}$, $u_4=-3$ and $u_5=2$ again, so the sequence has period 4. One full cycle sums to $2+\frac{1}{3}-\frac{1}{2}-3=-\frac{7}{6}$, and 100 terms make exactly 25 cycles, so the sum is $-\frac{175}{6}$. **Answer A.**

Question 5

Write $t=\log_a b$, so that $\log_b a=\frac{1}{t}$ and $t+\frac{1}{t}=\frac{10}{3}$, giving $t=3$ or $t=\frac{1}{3}$. Since $a>b>1$ we need $b=a^t<a$, so $t<1$ and $t=\frac{1}{3}$, that is $b=a^{1/3}$. Then $ab=a^{4/3}=256=2^8$, so $a=2^6=64$ and $b=4$. Hence $a-b=60$. The other root $t=3$ gives $(a,b)=(4,64)$, which has $a<b$ and is excluded. **Answer D.**

Question 6

Moving everything to one side and pairing, $\sin^2 x-\cos^2 x$ is $-\cos 2x$ and $\sin^2 2x-\cos^2 2x$ is $-\cos 4x$, so the equation is $\cos 2x+\cos 4x=0$. The sum-to-product identity turns this into $2\cos 3x\cos x=0$. Now $\cos x=0$ gives $x=90°,270°$, and $\cos 3x=0$ gives $x=30°,90°,150°,210°,270°,330°$. The two lists overlap at $90°$ and $270°$, so there are 6 solutions in total. **Answer D.**

Kevinvariant
kevinvariant08.github.io

Question 7

The polynomial $p(x) - x$ is cubic and vanishes at $x = 1, 2, 3$, so $p(x) - x = a(x-1)(x-2)(x-3)$ for some constant a. Putting $x = 4$ gives $5 - 4 = 6a$, so $a = \frac{1}{6}$. Then $p(5) = 5 + \frac{1}{6} \cdot 4 \cdot 3 \cdot 2 = 9$. **Answer C.**

Question 8

Substitute $u = x - 1$. Then $x^2 - 2x = (x-1)^2 - 1 = u^2 - 1$, and the equation reads $\left|u^2 - 1\right| = |u|$, so $u^2 - 1 = u$ or $u^2 - 1 = -u$. These are $u^2 - u - 1 = 0$ and $u^2 + u - 1 = 0$, each with discriminant 5, giving four distinct real values of u and hence four real solutions for x. **Answer E.**

Question 9

The ordered triples (a, b, c) of dice scores with $a+b+c = 10$ number 27: there are $\binom{9}{2} = 36$ ways to write 10 as an ordered sum of three positive integers, and $3 \times \binom{3}{2} = 9$ of these have a part exceeding 6. Among the 27, the triples with all scores different come from the sets $\{1, 3, 6\}$, $\{1, 4, 5\}$ and $\{2, 3, 5\}$, each in 6 orders, giving 18. The conditional probability is $\frac{18}{27} = \frac{2}{3}$. **Answer D.**

Question 10

The coefficient of x^2 in $(1+x)^k$ is $\binom{k}{2}$, so the required coefficient is $\sum_{k=2}^{20} \binom{k}{2} = \binom{21}{3} = 1330$ by the hockey-stick identity. Alternatively, the sum is a geometric series equal to $\frac{(1+x)^{21} - (1+x)^2}{x}$, and the coefficient of x^2 there is the coefficient of x^3 in $(1+x)^{21}$, namely $\binom{21}{3}$. **Answer B.**

Question 11

The gradient of the curve at P is $2a$, so the normal has gradient $-\frac{1}{2a}$ and equation $y - a^2 = -\frac{1}{2a}(x - a)$. Meeting this with $y = x^2$ gives $(x-a)(x+a) = -\frac{1}{2a}(x-a)$, and discarding the root $x = a$ leaves $x = -a - \frac{1}{2a}$ for Q. Its y-coordinate is $\left(a + \frac{1}{2a}\right)^2$, and by AM-GM $a + \frac{1}{2a} \geq 2\sqrt{\frac{1}{2}} = \sqrt{2}$, with equality at $a = \frac{1}{\sqrt{2}}$. The least y-coordinate is therefore 2. **Answer C.**

Question 12

The line meets the curve where $x^2 - 4 = 5$, so at $x = \pm 3$; the branch $4 - x^2 = 5$ has no solutions. The area is $\int_{-3}^{3} \left(5 - \left|x^2 - 4\right|\right) dx = 30 - \int_{-3}^{3} \left|x^2 - 4\right| dx$. By symmetry the last integral is $2\left(\int_0^2 (4 - x^2)\, dx + \int_2^3 (x^2 - 4)\, dx\right) = 2\left(\frac{16}{3} + \frac{7}{3}\right) = \frac{46}{3}$, so the area is $30 - \frac{46}{3} = \frac{44}{3}$. **Answer F.**

Question 13

For the original eight numbers, $\sum x_i = 40$ and $\sum x_i^2 = 8\left(4 + 5^2\right) = 232$. Removing 1 and 9 leaves six numbers with $\sum x_i = 30$ and $\sum x_i^2 = 232 - 1 - 81 = 150$. Their mean is still 5 and their variance is $\frac{150}{6} - 25 = 0$. In other words the six remaining numbers are all equal to 5; the data set was forced to be $1, 5, 5, 5, 5, 5, 5, 9$ all along. **Answer A.**

Question 14

We need $x \geq 1$; put $t = \sqrt{x-1} \geq 0$, so $x = t^2 + 1$. Then $x + 3 - 4\sqrt{x-1} = t^2 - 4t + 4 = (t-2)^2$ and $x + 8 - 6\sqrt{x-1} = (t-3)^2$, so the equation is $|t-2| + |t-3| = 1$. The sum of distances from t to 2 and to 3 equals 1 exactly when t lies between them, so $2 \leq t \leq 3$, that is $5 \leq x \leq 10$. Every x in that interval works. **Answer F.**

Kevinvariant
kevinvariant08.github.io

Question 15

For $k = 1, 2, \ldots, 9$, the value $\lfloor \sqrt{n} \rfloor$ equals k for $k^2 \leq n \leq (k+1)^2 - 1$, which is $2k+1$ values of n. These contribute $\sum_{k=1}^{9} k(2k+1) = 2\sum k^2 + \sum k = 2 \cdot 285 + 45 = 615$. The term $n = 100$ contributes 10, giving 625. **Answer B.**

Question 16

Let $y = \dfrac{\sin x}{2 + \cos x}$, so $\sin x - y\cos x = 2y$. The left-hand side has the form $R\sin(x - \alpha)$ with $R = \sqrt{1 + y^2}$, so it can only equal $2y$ if $4y^2 \leq 1 + y^2$, that is $y^2 \leq \frac{1}{3}$. The value $y = \frac{1}{\sqrt{3}}$ is attained, since with this y the equation $\sin(x - \alpha) = \frac{2y}{R} = 1$ is solvable. The maximum is $\frac{\sqrt{3}}{3}$. **Answer D.**

Question 17

Using $\cos 2x = 1 - 2\sin^2 x$, the inequality becomes $2\sin^2 x + \sin x - 1 \geq 0$, which factorises as $(2\sin x - 1)(\sin x + 1) \geq 0$. The factor $\sin x + 1$ is never negative, so the product is non-negative exactly when $\sin x \geq \frac{1}{2}$ or $\sin x = -1$. On $[0, 2\pi]$ the first gives $\frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$ and the second gives the single point $x = \frac{3\pi}{2}$. Forgetting the isolated point is the common slip. **Answer B.**

Question 18

Let d be the perpendicular distance from O to the line. Since the line passes through P, which is at distance 1 from O, we have $0 \leq d \leq 1$. The chord has length $2\sqrt{4 - d^2}$, so the area is $d\sqrt{4 - d^2} = \sqrt{d^2(4 - d^2)}$. As a function of d^2 this is a downward parabola with vertex at $d^2 = 2$, so on $0 \leq d \leq 1$ it is increasing and the maximum is at $d = 1$, giving area $\sqrt{3}$. The tempting answer 2 would require a right angle at O, which needs $d = \sqrt{2}$, out of reach. **Answer C.**

Question 19

If ℓ has equation $y = mx + c$ and touches the curve at $x = p$ and $x = q$, then $x^4 - 4x^3 + 2x^2 - mx - c = (x - p)^2(x - q)^2$ identically. Expanding the right-hand side and comparing coefficients of x^3 gives $p + q = 2$, and comparing x^2 gives $p^2 + 4pq + q^2 = 2$, so $(p + q)^2 + 2pq = 2$ and $pq = -1$. The coefficient of x on the right is $-2pq(p + q) = 4$, so $-m = 4$ and $m = -4$. For completeness, $c = -p^2q^2 = -1$, and p, q are the real roots of $t^2 - 2t - 1 = 0$, so the line $y = -4x - 1$ does exist. **Answer A.**

Question 20

Since $f(2) = 10$ and f is increasing, f^{-1} maps $[0, 10]$ onto $[0, 2]$. The graph of f^{-1} is the reflection of the graph of f in $y = x$, so the area under f^{-1} from 0 to 10 equals the area between the graph of f and the y-axis for $0 \leq y \leq 10$, which is the 2×10 rectangle minus the area under f from 0 to 2. Hence $\int_0^{10} f^{-1}(x)\,dx = 20 - \int_0^2 \left(x^3 + x\right) dx = 20 - (4 + 2) = 14$. **Answer E.**

Kevinvariant
kevinvariant08.github.io